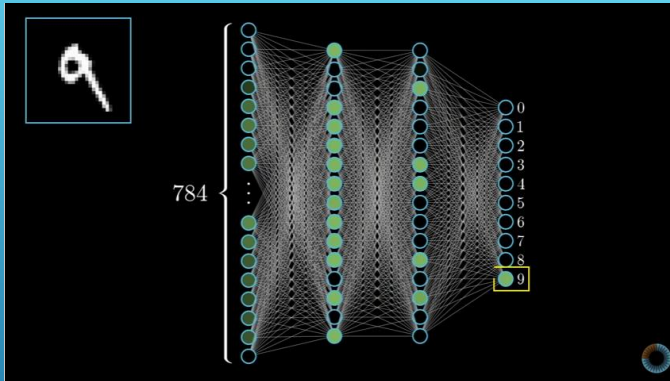


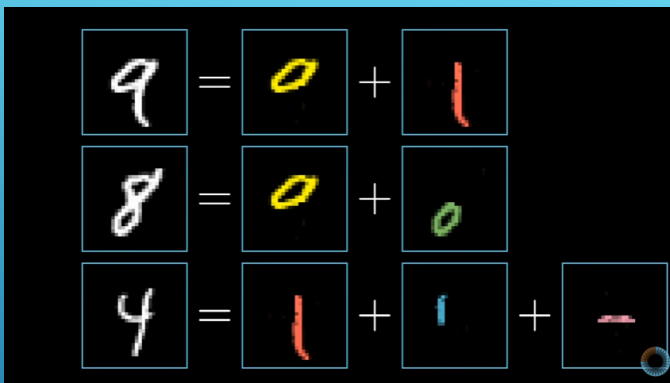
CONVOLUTIONAL NEURAL NETWORKS

Deep learning

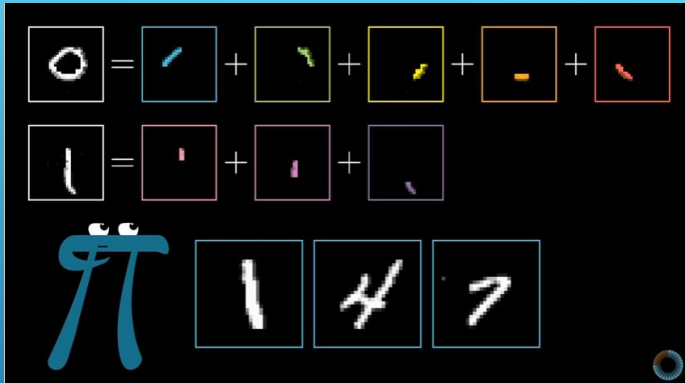




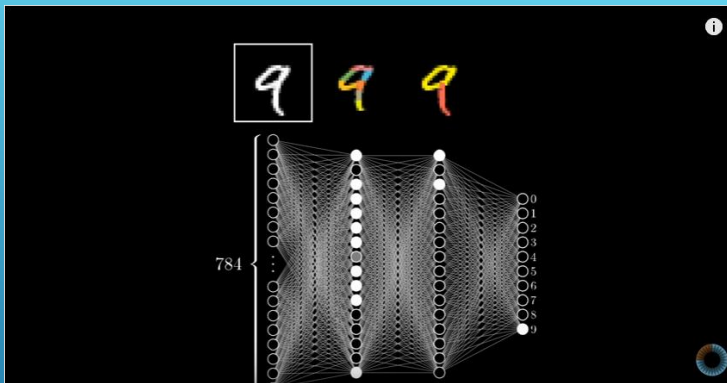
INTUITION BEHIND MULTILAYER NETWORKS



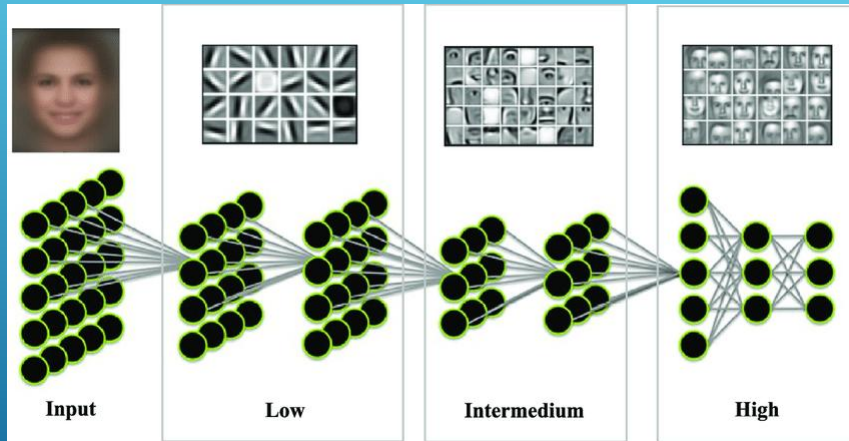
INTUITION BEHIND MULTILAYER NETWORKS



INTUITION BEHIND MULTILAYER
NETWORKS



INTUITION BEHIND MULTILAYER
NETWORKS



A toy ConvNet: X's and O's

Says whether a picture is of an X or an O

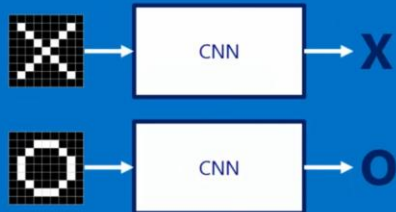
A two-dimensional
array of pixels



X or **O**

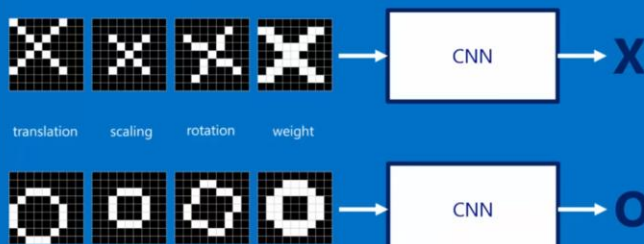
CONVOLUTIONAL NEURAL NETWORKS

For example



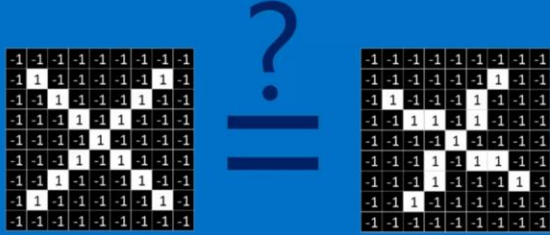
EXAMPLE

Trickier cases

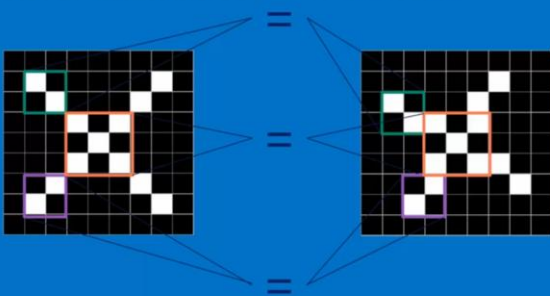


DIFFERENT CASES

What computers see



ConvNets match pieces of the image



Features match pieces of the image

1	-1	-1
-1	1	-1
-1	-1	1

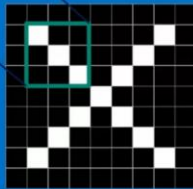
1	-1	1
-1	1	-1
1	-1	1

-1	-1	1
-1	1	-1
1	-1	-1

1	-1	-1
-1	1	-1
-1	-1	1

1	-1	1
-1	1	-1
1	-1	1

-1	-1	1
-1	1	-1
1	-1	-1



Filtering: The math behind the match

1	-1	-1
-1	1	-1
-1	-1	1

-1	-1	-1	-1	-1	-1	-1	-1
-1	1	-1	-1	-1	-1	1	-1
-1	-1	1	-1	-1	-1	-1	-1
-1	-1	-1	1	-1	-1	-1	-1
-1	-1	-1	-1	1	-1	-1	-1
-1	-1	1	-1	-1	-1	-1	-1
-1	1	-1	-1	-1	-1	1	-1
-1	-1	-1	-1	-1	-1	-1	-1

FILTERING

Filtering: The math behind the match

1. Line up the feature and the image patch.
2. Multiply each image pixel by the corresponding feature pixel.
3. Add them up.
4. Divide by the total number of pixels in the feature.

Play (N)

FILTERING

Filtering: The math behind the match

$$\begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 1 \end{bmatrix} \times \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 1$$

1	1	1
1	1	1
1	1	1

-1	-1	-1	1	-1	-1	-1	-1
-1	1	-1	-1	-1	-1	1	-1
-1	-1	1	-1	-1	1	-1	-1
-1	-1	-1	1	1	-1	-1	-1
-1	-1	-1	-1	1	-1	-1	-1
-1	-1	-1	-1	-1	1	-1	-1
-1	-1	1	-1	-1	-1	1	-1
-1	1	-1	-1	-1	-1	-1	1
-1	-1	-1	-1	-1	-1	-1	-1

FILTERING

Filtering: The math behind the match

1	-1	-1
-1	1	-1
-1	-1	1

1	1	1
1	1	1
1	1	1

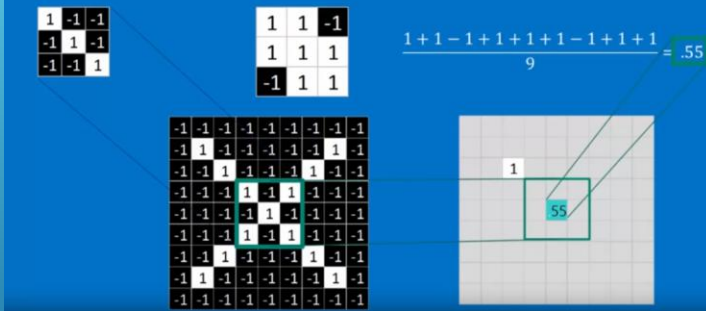
$$\frac{1+1+1+1+1+1+1+1+1}{9} = 1$$

-1	-1	-1	-1	-1	-1	-1	-1
-1	1	-1	-1	-1	-1	1	-1
-1	-1	1	-1	-1	1	-1	-1
-1	-1	-1	1	1	-1	-1	-1
-1	-1	-1	-1	1	-1	-1	-1
-1	-1	1	-1	-1	-1	1	-1
-1	1	-1	-1	-1	-1	-1	1
-1	-1	-1	-1	-1	-1	-1	-1

1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1

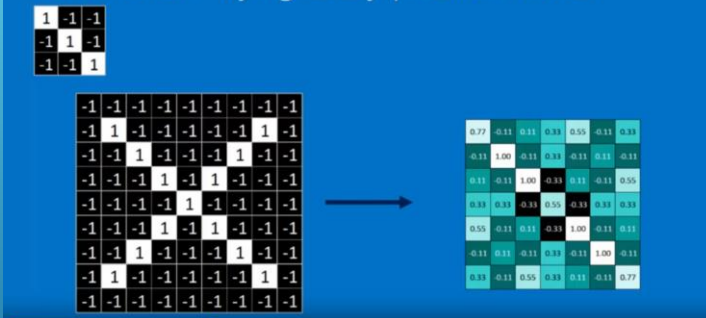
FILTERING

Filtering: The math behind the match



FILTERING

Convolution: Trying every possible match

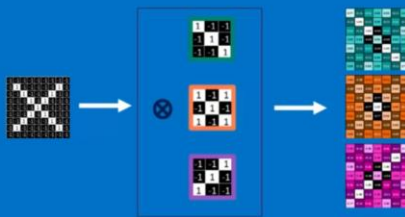


CONVOLUTION



CONVOLUTION

Convolution layer
One image becomes a stack of filtered images



CONVOLUTION

Pooling: Shrinking the image stack

1. Pick a window size (usually 2 or 3).
2. Pick a stride (usually 2).
3. Walk your window across your filtered images.
4. From each window, take the maximum value.

Play (x)

POOLING

Pooling

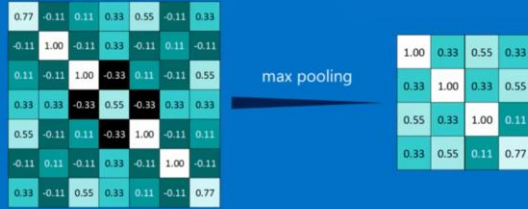
0.77	-0.11	0.11	0.33	0.55	-0.11	0.33
-0.11	1.00	-0.11	0.33	-0.11	0.11	-0.11
0.11	-0.11	1.00	-0.33	0.11	-0.11	0.55
0.33	0.33	-0.33	0.55	-0.33	0.33	0.33
0.55	-0.11	0.11	-0.33	1.00	-0.11	0.11
-0.11	0.11	-0.11	0.33	-0.11	1.00	-0.11
0.33	-0.11	0.55	0.33	0.11	-0.11	0.77

maximum

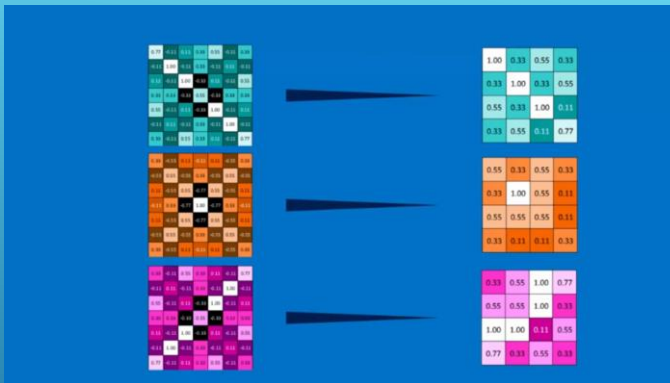
1.00			

POOLING

Pooling



POOLING



POOLING

Normalization

Keep the math from breaking by tweaking each of the values just a bit.

Change everything negative to zero.

NORMALIZATION

Rectified Linear Units (ReLUs)

0.77	-0.11	0.11	0.33	0.55	-0.11	0.33
-0.11	1.00	-0.11	0.33	-0.11	0.11	-0.11
0.11	-0.11	1.00	-0.33	0.11	-0.11	0.55
0.33	0.33	-0.33	0.55	-0.33	0.33	0.33
0.55	-0.11	0.11	-0.33	1.00	-0.11	0.11
-0.11	0.11	-0.11	0.33	-0.11	1.00	-0.11
0.33	-0.11	0.55	0.33	0.11	-0.11	0.77



0.77	0	0.11	0.33	0.55	0	0.33
0	1.00	0	0.33	0	0.11	0
0.11	0	1.00	0	0.11	0	0.55
0.33	0.33	0	0.55	0	0.33	0.33
0.55	0	0.11	0	1.00	0	0.11
0	0.11	0	0.33	0	1.00	0
0.33	0	0.55	0.33	0.11	0	0.77

RELU

ReLU layer

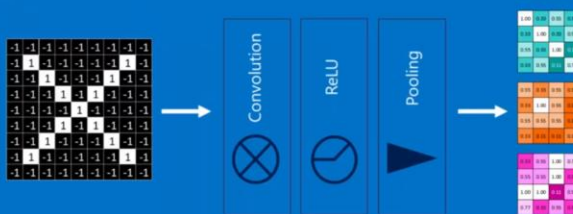
A stack of images becomes a stack of images with no negative values.



RELU

Layers get stacked

The output of one becomes the input of the next.



STACKING LAYERS

Deep stacking

Layers can be repeated several (or many) times.



DEEP STACKING

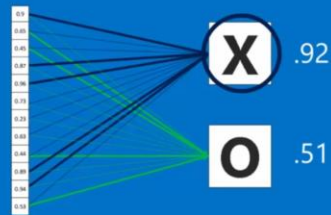
Fully connected layer

Vote depends on how strongly a value predicts X or O



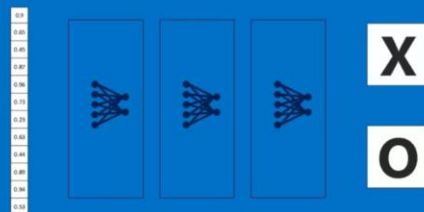
FULLY CONNECTED LAYERS

Fully connected layer
Future values vote on X or O



FULLY CONNECTED LAYERS

Fully connected layer
These can also be stacked.



FULLY CONNECTED LAYERS

Putting it all together

A set of pixels becomes a set of votes.



HOW TO VISUALIZE A CONVOLUTIONAL NEURAL NETWORK?

Learning

Q: Where do all the magic numbers come from?

Features in convolutional layers

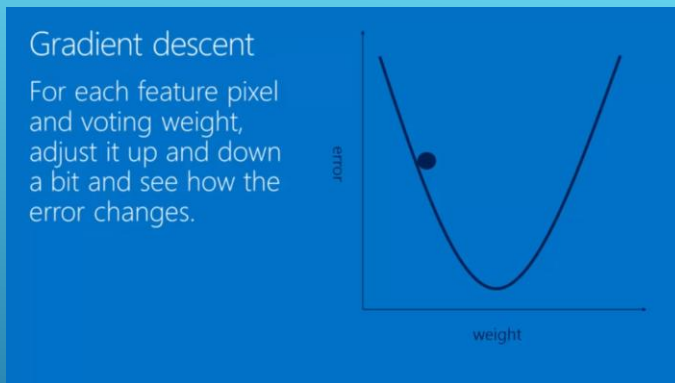
Voting weights in fully connected layers

A: Backpropagation

WHERE DO THE WEIGHTS COME FROM?



BACKPROPAGATION



GRADIENT DESCENT

Hyperparameters (knobs)

Convolution

Number of features

Size of features

Pooling

Window size

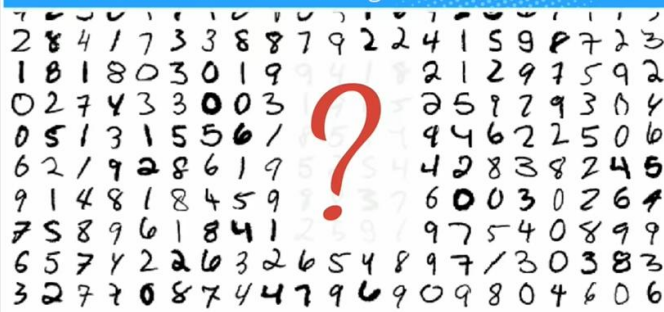
Window stride

Fully Connected

Number of neurons

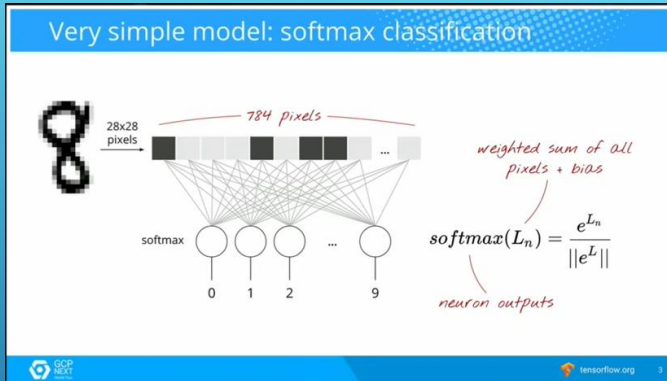
HYPERPARAMETERS

Hello World: handwritten digits classification - MNIST

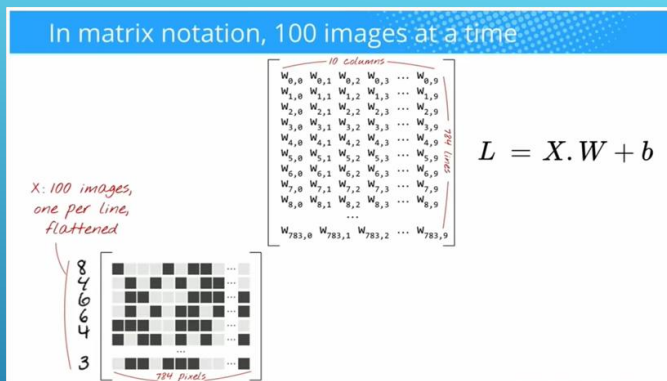


MNIST = Mixed National Institute of Standards and Technology - Download the dataset at <http://yann.lecun.com/exdb/mnist/>

HANDWRITTEN DIGIT RECOGNITION

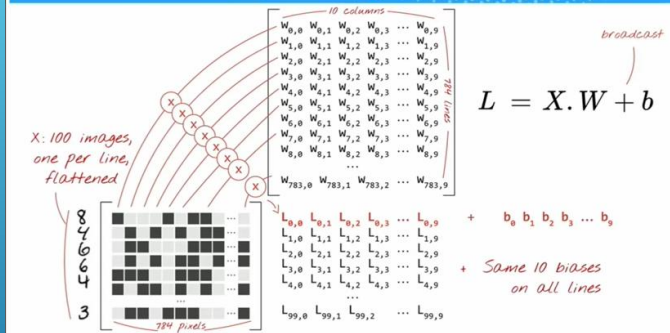


NEURAL NETWORKS WITH SOFTMAX

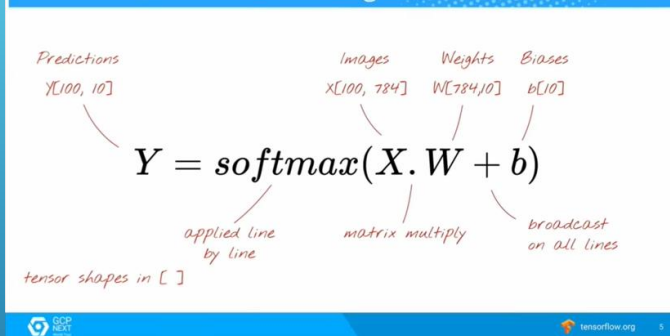


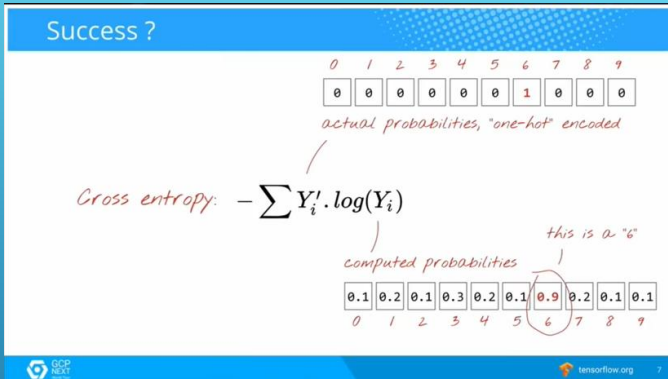
USING A SET OF IMAGES

In matrix notation, 100 images at a time

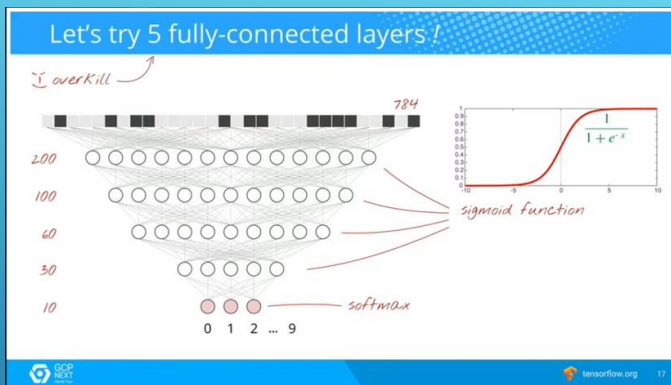


Softmax, on a batch of images





CROSS ENTROPY FOR LOSS CALCULATION



MULTILAYER DEEP NEURAL NETWORKS

- ▶ Gradient descent
- ▶ Backpropagation
- ▶ Activation functions
 - ▶ Softmax
 - ▶ Sigmoid
 - ▶ ReLU
- ▶ Other architectures
 - ▶ *Long-Short Term Memory* (LSTM) and recurrent neural networks
 - ▶ Autoencoders

IMPORTANT CONCEPTS

- ▶ How Convolutional Neural Networks work.
<https://www.youtube.com/watch?v=FmpDIaiMleA> (consultado el 17 de febrero de 2020)
- ▶ Tensorflow and deep learning - without a PhD by Martin Görner.
<https://www.youtube.com/watch?v=vq2nnJ4g6N0> (consultado el 17 de febrero de 2020)

REFERENCES